

CSI132 - DISCRETE STRUCTURES II TUTORIAL
ATTEMPT ALL QUESTIONS
DURATION: 2 HOURS
SUMMATIONS AND ~~MATHEMATICAL INDUCTION~~

1. Write the following using summation symbol(s).

(a) $a_3 + a_4 + a_5 + \dots + a_{n-1}$

(b) $a_2 + a_4 + a_6 + a_8 + a_{10} + \dots + a_{2n}$

(c) $a_3 + a_5 + a_7 + \dots$

(d) $c_9 + c_8 + c_7 + \dots + c_2 + c_1$

(e) $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$

(f) $n + (n-1) + (n-2) + \dots + 2 + 1$

(g) $-1 + 2 - 3 + 4 - 5 + \dots$

(h) $a_i + a_{i+2} + a_{i+4} + a_{i+6} + a_{i+8} + \dots + a_{i+2n}$

(i) $\frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \dots + \frac{n+1}{2n}$

2. Given that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, how can you evaluate $\sum_{i=34}^n i$?

3. Evaluate $\sum_{j=40}^{100} j$

4. Evaluate $\sum_{10}^{1000} 3$

5. Expand $\sum_{i=0}^n \frac{(-1)^i}{i+1}$

6. Expand and simplify $\sum_{i=1}^5 (a_i - a_{i-1})$

7. Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the last term.

8. Rewrite $\sum_{k=0}^n 2^k + 2^{n+1}$ as a single summation.

9. We sometimes can simplify summation by changing the variable of summation. For example, suppose $\sum_{k=0}^6 \frac{1}{k+1}$. Let us make the variable change, $j = k + 1$. Basically we need to replace $k + 1$ or k whenever they occur. Systematically, we compute first the lower bound of the summation, followed by the upper bound of the summation, and lastly the term of the summation. In our case the lower bound is $k = 0$, hence in that case $j = k + 1 = 0 + 1 = 1$; the upper bound is $k = 6$, and therefor the new upper bound is $j = k + 1 = 6 + 1 = 7$. The term $\frac{1}{k+1}$ becomes $\frac{1}{j}$. Hence we have $\sum_{k=0}^6 \frac{1}{k+1} = \sum_{j=1}^7 \frac{1}{j}$. The new summation looks a lot better.

- (a) Express $\sum_{i=1}^{n-1} \frac{i}{(n-i)^2}$ as a new summation by using $j = i - 1$ to change the summation variable.
- (b) Express $\sum_{i=3}^n \frac{i}{i+n-1}$ as a new summation by using $j = i - 1$ to change the summation variable.
10. Let $a_1 = -2$, $a_2 = -1$, $a_3 = 0$, $a_4 = 1$, and $a_5 = 2$. Compute the following:
- (a) $\sum_{i=1}^5 a_i$
- (b) $\sum_{i=0}^2 a_{2i+1}$
11. Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the last three terms.
12. Rewrite $\sum_{k=0}^n 2^k$ by separating off the first and the last terms.
13. For each integer n with $n \geq 1$, let $P(n)$ be the formula:
- $$1 + 3 + 5 + \dots + (2n - 1) = n^2$$
- (a) Write $1 + 3 + 5 + \dots + (2n - 1)$ using a summation symbol.
- (b) Using mathematical induction, prove that: summation you derived above $= n^2$, $\forall n \geq 1$. Use the template we used in *CSI131*.
14. Using mathematical induction, prove that: $\sum_{i=1}^n (3i - 2) = \frac{n(3n-1)}{2}$, $\forall n \geq 1$. Use the template we used in *CSI131*.
15. Using mathematical induction, prove that: $\sum_{i=1}^n \left(\frac{1}{i(i+1)}\right) = \frac{n}{n+1}$, $\forall n \geq 1$. Use the template we used in *CSI131*.